

Multidimensional Scaling (Intro)

Purpose

- **Visualization of clustering for high-dimensional ($p > 3$) data**
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Input

- $(n \times p)$ data matrix, $\mathbf{X}$, with n p -dimensional multivariate observations is used to calculate $(n \times n)$ symmetrical zero-diagonal distance matrix, D , where entry $d_{ij}^{(p)}$ is the distance (Euclidean or other) in p dimensions between the observations i and j
 - Alternatively, the distance matrix, D , is directly the input to the algorithm (multidimensional scaling can create “maps” from distances)
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Classical metric multidimensional scaling

- Mapping and plot of the n p -dimensional observations in $q < p$ dimensions (usually $q = 2$ or $q = 3$) in such a way that the metric distances between observations are approximately preserved in the q -dimensional plot, i.e. $d_{ij}^{(p)} \approx d_{ij}^{(q)}$ for all i, j
 - Usually, a perfect match of all p -dimensional distances is not possible in $q < p$ dimensions
 - The deviations are measured by a stress function (cost function) which then can be minimized (optimization problem)
 - Many stress functions can be proposed (see, e.g., the textbook), but basically they all try to minimize the distance errors $|d_{ij}^{(p)} - d_{ij}^{(q)}|$
 - Note that multidimensional scaling is not a simple projection from p to q dimensions (since the latter is just deleting $p-q$ coordinates in the observations)
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Generalizations

- Multidimensional scaling can also be used for datasets where Euclidean or other numerical distance measures between observations are not meaningful, e.g. categorical or non-numerical variables
 - In this case, only the rank order of the “distances” or “similarities” between observations is considered in the p - to q -dimensional transformation
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Connection with PCA

- Classical metric multidimensional scaling from p to q dimensions using Euclidean distance is equivalent to plotting the first q PCA-scores for each observation (possibly with reflection (change of sign) of axes), if the principal components are obtained from the sample covariance matrix of the observations
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